

EMY: Supporting Autism Therapy with a Socially Assistive Robot

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Figure 1: From left to right: Emy robot displaying different facial expressions; setup for the therapeutic sessions including a screen between the robot and the child; example of the Animals game interface.

Abstract

We present Emy, a low-cost, socially assistive robot designed to support therapeutic sessions for children with Autism Spectrum Disorder (ASD) within the EMOROB CARE project. We detail the robot's multimodal interaction capabilities, including a comprehensive set of facial expressions and reactive behaviours, as well as its perception and speech interaction system. The demonstration showcases the robot together with a suite of four therapeutic games that promote cognitive and social skills. The system's modularity and affordability increase its accessibility for home-based ASD interventions.

CCS Concepts

• **Human-centered computing**; • **Computer systems organization** → **Robotics**; • **Computing methodologies** → *Artificial intelligence*;

Keywords

Socially Assistive Robots (SAR), Autism Spectrum Disorder (ASD), Low-Cost Robotics, Multimodal Interaction, Therapeutic Games

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1 Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects social communication, interaction, and engagement. Traditional therapy methods face several challenges, including limited therapist availability, difficulty engaging children consistently, and a persistent need for individualized, adaptive interventions [5].

Socially Assistive Robots (SAR) are emerging as a promising tool to support these therapies by providing structured, predictable, and safe interactions, which is particularly beneficial for children with ASD [2, 4]. However, many existing therapeutic robots (e.g., NAO, QTrobot, MiRo) are often expensive and not accessible for widespread home use, limiting their application beyond clinical settings [3].

The EMOROB CARE project [1] addresses this challenge by developing a modular, low-cost, and interactive social robot to support therapeutic sessions for children with ASD. This paper and accompanying video present its architecture and multimodal interactive features for therapeutic games, with contributions lying primarily in system integration, interaction design, and therapeutic orchestration of existing AI and robotics modules rather than in new perception or language algorithms.

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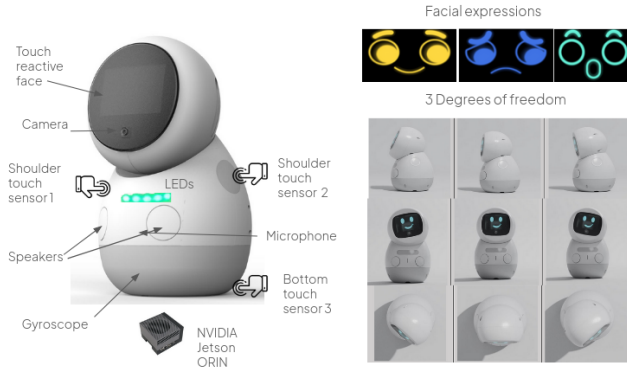


Figure 2: The EMOROB CARE robot hardware, illustrating its compact design, sensors and expressive face and head DoF.

2 Robot Design

The robot design prioritizes a compact structure (20–30 cm) that is stable and ergonomic. The overall system is divided into physical robot hardware and a multi-component software architecture.

2.1 The Hardware

The robot’s physical structure includes an expressive face, a head with 3 Degrees of Freedom (DoF) for expressive movement, and various sensors (Figure 2). Specifically, a touch reactive face, a camera, 2 speakers, and multi-coloured LEDs on the frontal side of the body. The robot is also equipped with multiple touch sensors located on the left shoulder, right shoulder, and the bottom of the unit, enabling touch interaction. The main processing unit for the robot is an NVIDIA Jetson ORIN Nano.

2.2 The Software

EMOROB CARE uses state-of-the-art open-source components (ASR, TTS, LLMs, and vision models), while novel development focuses on their orchestration, real-time interaction management, and therapeutic task control through the Mission Controller. The architecture (Figure 3) comprises three blocks: a perception system (blue), a speech and communication system (red), and a behavioural system (orange) centred on the Mission Controller. Game content is rendered on an external HTML/CSS interface. An idle behaviour runs when the robot is not in a game. Hardware modules (grey) provide sensing and actuation. Interaction data are stored in an SQLite database. ASR, TTS, and LLMs run on an external workstation for performance, while the robot handles all other components and physical outputs.

3 Robot Interaction Modalities

The robot is designed for multi-modal interaction, using tactile, visual, and verbal inputs to create an engaging and adaptive therapeutic experience.

3.1 Expressiveness and Idle Behaviour

The robot’s core functionality includes expressive communication and responsive idle behaviour. It can display 12 distinct facial expressions: Neutral, Happy, Sad, Love, Surprise, Angry, Wink, Bored, Sceptical, Thinking, Waiting, and Sleepy.

The robot’s state management is defined by the following logic:

- **Initial State:** The robot is sleeping when first switched on.
- **Wake-up:** Touching any body or face sensor, or using the verbal "greet" topic, transitions the robot to the Neutral expression, and it greets the user.
- **Sleep:** A "goodbye" topic command returns the robot to Sleepy mode, where it will not react to stimuli.

When the robot is awake, it exhibits several dynamic reactions:

- **Face Touch:** It blinks constantly when any part of the face is touched.
- **Shoulder Touch:** Touching the left_shoulder or right_shoulder (or both) triggers a Love reaction.
- **Verbal Input:** Detecting keywords via the ASR calls the chitchat LLM to generate a contextual response. Sentences related to predefined topics are handled with specific, appropriate answers.
- **Movement:** If the robot is moved (e.g., lifted or shaken), it displays a Surprise reaction.

3.2 Visual Perception

The robot’s vision system is built on existing deep-learning modules for face, body, gesture, and object recognition, which are integrated into a unified perception pipeline for therapeutic interaction.

Face identification, built on off-the-shelf recognition models, is a critical feature, allowing the robot to recognize the therapist and the child, and enabling adaptation of the robot behaviour based on the child’s established patterns. Furthermore, engagement detection monitors where the child’s attention is focused (robot, external tablet, or therapist) to detect distraction and help recover attention during a session. The system also recognizes yes/no and pointing gestures as important communication tools for autistic children.

3.3 Verbal Interaction

Verbal interaction relies on existing pre-trained Automatic Speech Recognition (ASR), Text-To-Speech (TTS), and Dialogue Management (DL) modules, integrated through a custom dialogue and task-control layer.

- **Automatic Speech Recognition (ASR):** The system uses Vosk models for Spanish language recognition. An external microphone is used to mitigate hardware constraints and address common anomalies in children’s speech, such as low volume or poor pronunciation.
- **Text-to-Speech (TTS):** The robot utilizes XTTS v2¹, an open-source model that supports Spanish and offers emotional cloning. This allows for fine-grained control over the robot’s emotion, tone, and rhythm, ensuring the voice is consistent across different emotional states.

¹<https://huggingface.co/coqui/XTTS-v2>

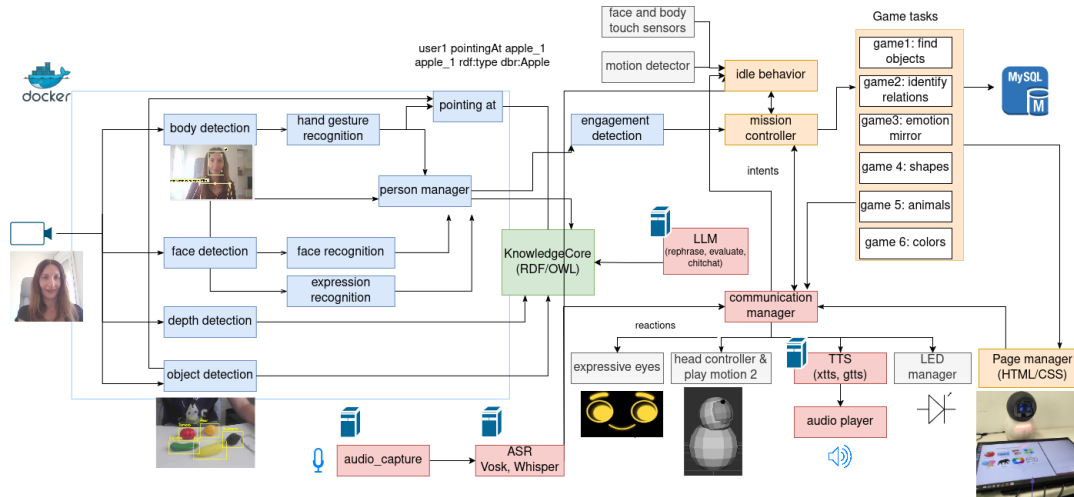


Figure 3: The EMOROB CARE software architecture: perception subsystem in blue; speech subsystem in red; behavioural subsystem in orange; and sensors and actuation modules in grey.

- **Dialogue:** The system employs various LLMs for three primary functions: utterance rephrasing to add variability to dialogue; evaluation of answers to classify the child’s response; and chitchat, to maintain the child’s attention between structured activities.

4 Therapeutic Activities

The novel contribution of this project is its suite of therapeutic games and their real-time orchestration through the robot’s multi-modal behaviour, to support therapists during their sessions:

- **Expressions:** the goal is to recognize, name, and request emotions based on the robot’s facial expressions. *Flow:* The system displays expressions; the child responds by verbally naming the emotion or selecting it in the screen.
- **Colours:** the objective is to recognize, name, and request colours using the robot’s lights and face. *Flow:* The system displays colours (via robot lights/face); the child responds verbally or selects the corresponding option.
- **Shapes:** the aim is to recognize, name, and trace different geometric shapes. *Flow:* The system shows geometric shapes and asks the child to identify or draw them.
- **Animals** the objective is to recognize animals and confirm if they match the robot’s prompt. *Flow:* The system shows images of animals and asks the child to confirm if the animal matches the prompt.

5 Demo Overview

The demo set-up requires the EMOROB CARE robot to be connected to a Wi-Fi hotspot, which will allow it to connect to the remote workstation hosting the speech pipeline. A laptop and a tablet to run the demo and display game content must be connected to the same Wi-Fi. Finally, an external microphone will be used to mitigate risk of misunderstanding due to event noise. The accompanying video shows what will be shown during the demo describing the robot’s dynamic expressive capabilities, including the Sleepy/Neutral

transition and various touch-based (e.g., Love on shoulder touch) or perceptual (e.g., gesture and face/object identification) reactions. The video also shows excerpts of two therapeutic games where the robot guides the user playing the various activities through voice and visual prompts on the external touch screen. The primary significance of the EMOROB CARE system lies in its low-cost, split-architecture design (Figure 3), a critical step towards the use of socially assistive robotics for ASD interventions. The system’s novelty is in its comprehensive, clinically aligned suite of games and its highly expressive, predictable, and engaging social behaviour.

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